

Supplemental Material

I. Handout given to students at the start of the lab (sample)

Exploring the electronic structure of carminic acid: an open-ended laboratory integrating art, commerce, and chemistry

Aim: To characterize carminic acid or its metal derivatives using available spectroscopic or analytical methods

Overview: Over the next four weeks, you will design and carry out an experiment to probe the electronic structure of carminic acid or one of its metal derivatives. Carminic acid is a glycosidic hydroxyanthraquinone that has been widely used as a colorant for paints and foods. It is extracted from a type of lice (called cochineal) that grow on cactus. You will be provided with as many grams of cochineal as you want and you have access to any of the instruments you have used this semester (UV-vis spectrophotometer, Raman spectrophotometer, FTIR spectrophotometer) as well as other potentially useful instruments in the department (potentiostat, UV lamp, NMR spectrophotometer). You can also use any routine glassware and supplies in the department. Each of you, working alone or with a partner, will choose a particular experiment and then take responsibility for completing it. Keep in mind the project goal: to understand something about the electronic structure of carminic acid or one of its metal complexes. By the end of the module, we will write a single lab report, in the style of a journal article, presenting our data and analyzing its significance.

Module structure: We have set up a shared computer folder that contains copies of a number of helpful references. We will ask you to share your work with the class in this shared folder and have set up weekly group meetings for us to discuss the progress we have made and our plans for the upcoming week. The first week of the module, you will search and review the literature and develop a procedure for extracting carminic acid from cochineal and decide what characterization methods you want to use. You will present this information at our first group meeting. You will have two full weeks (4 four-hour lab periods) to extract the carminic acid and make your intended measurements. You are welcome to divide up the initial work – e.g. not every single group needs to carry out the extraction but make sure that you intercept the material at the stage needed for your measurements. At the end of the third week, each of you will present your results and preliminary conclusions to the group where they can be discussed. If any additional experiments need to be done, we can plan for that to occur. The final week of the module will be spent writing the final report and preparing figures. Keep in mind all that you have learned about the proper scientific writing style, data analysis and data presentation as you prepare your contributions to the final report. We will spend the last two days of the module editing the work of our classmates so that we end with a document we all admire.

II. Materials and methods

All procedures reported in this experiment were carried out by at least two independent teams of students and independently by at least one faculty member. These are methods that work. We did not present them to the students initially but different instructors might have different views about how much literature work was important to their curricular goals.

Chemical name	Formula	Catalog Number	Supplier	CAS Number
Carminic acid	$C_{22}H_{20}O_{13}$	1.00211.0005	EMD Millipore Corporation	1260-17-9
Carmine	$Ca(C_{22}H_{18}O_{13})_2AlOH \cdot 3H_2O$	190200050	Acros	1390-65-4
Sodium carbonate	Na_2CO_3	S263-500	Fisher Scientific	497-19-8
Ammonium hydroxide	NH_4OH	A669-212	Fisher Scientific	7664-41-7
Hydrochloric acid	HCl	A144C-212	Fisher Scientific	7647-01-0
Yttrium nitrate	$Y(NO_3)_3 \cdot 6H_2O$	23795-7	Aldrich	13494-98-9
Tetrabutylammonium hexafluorophosphate	$(CH_3CH_2CH_2CH_2)_4N(PF_6)$	86879	Sigma-Aldrich	309-63-5
Cochenille (cochineal)		36040	Kremer Pigment	

Table 1. Chemicals and materials

Instrument	Company	Parameters
Cary 50 Bio UV-Vis spectrophotometer	Varian	Avg. time: 0.1 s Data interval: 1.0 nm
PC1 Photon counting fluorimeter	ISS	Excitation wavelength: 490 nm - 570 nm Emission wavelength: 500 nm - 800 nm Scan rate: 1.0 nm/s
Spectrum 100 FT-IR spectrometer	Perkin Elmer	Resolution: 4 cm^{-1}
μ Auto III Potentiostat /Galvanostat	Metrohm USA	Voltage range: 0.3-0.85 V Number of cycles: 5
AB 15 pH Meter	Fisher Scientific	
150 Watt Xenon lamp/power supply Model 68805	DREL Instrument	
Universal ARC lamp Housing Model	DREL Instrument	
Grating monochromator Model 77250	Thermo Oriel	

Table 2. Instruments and apparatus

1. Carmine synthesis

1.1 Carminic acid extraction

0.24 g cochineal was ground to a powder with a mortar and pestle. Sieving was not necessary because of the high quality of the cochineal used but cochineal from other sources might need to be sieved to remove solid impurities that would degrade from chromophore intensity. The ground cochineal was placed in a 600-ml beaker to which 300 ml of deionized (DI) water was added. Alternative procedures for carminic acid isolation recommend using 0.1 M sodium carbonate or 0.1 M ammonium hydroxide solution at this step. In our hands the carminic acid produced with these extraction methods less closely matched the authentic purchased standard. Boiling chips were added to the solution, which was boiled for 15-20 min. The purple-red solution was filtered with a fritted Büchner funnel and stored in an amber bottle. A Cary 50 Bio UV-Vis spectrophotometer was used to measure the UV-Vis spectrum of carminic acid. 1.00 ml of the filtrate was transferred to a 1.5 ml-microcentrifuge tube. Water was removed by lyophilization. A Spectrum 100 FT-IR spectrometer was used to measure the IR spectrum of the powdered carminic acid.

To make quantitative solutions of well-defined pH, the following procedures were used. 0.1000 g of carminic acid were dissolved in MilliQ water and transferred to a 100-mL volumetric flask and diluted to the mark to make a 0.02 mM carminic acid stock solution. 10.00 mL of this stock solution was pipetted into a 100-mL volumetric flask and diluted to the mark to make a solution with a measured pH of 3.46.

To make carminic acid solutions of pH 2.0, 6.0, 10.1 and 12.0 respectively, 10.00 mL of the stock solution was pipetted into a 150-mL beaker, to which 80 mL of MilliQ water was added. These solutions were titrated with 0.1 M HCl or 0.1 M NaOH to pH 2.0, 6.0, 10.0, or 12.0 respectively. Once the target pH was reached, the solutions were transferred to 100-mL volumetric flasks and diluted to the mark with MilliQ water. These solutions were used to obtain UV-Vis and fluorescence spectra and to make electrochemical measurements.

1.2 Carmine synthesis

9 mL of carminic acid solution as extracted was placed in a beaker to which 1.0 M citric acid and 0.5 M aluminum potassium sulfate solution were added simultaneously dropwise until pH 5.5 was reached. During addition, the solution changes color from purple to reddish purple. Once pH 5.5 was reached, the solution was boiled for 5 min and then filtered. The product was rinsed repeatedly with room temperature DI water until the filtrate was colorless. The solid product, carmine, was air dried and stored.

2. Characterization of carmine and carminic acid

Carminic acid was characterized by IR spectroscopy, UV-Vis spectroscopy, fluorescence spectroscopy, and electrochemistry. The stability of carminic acid to photochemical degradation was studied by irradiating with a 150 W Xenon lamp. The stoichiometry of the reaction between carminic acid and yttrium was studied using Job's method.

2.1 UV-Visible spectroscopy

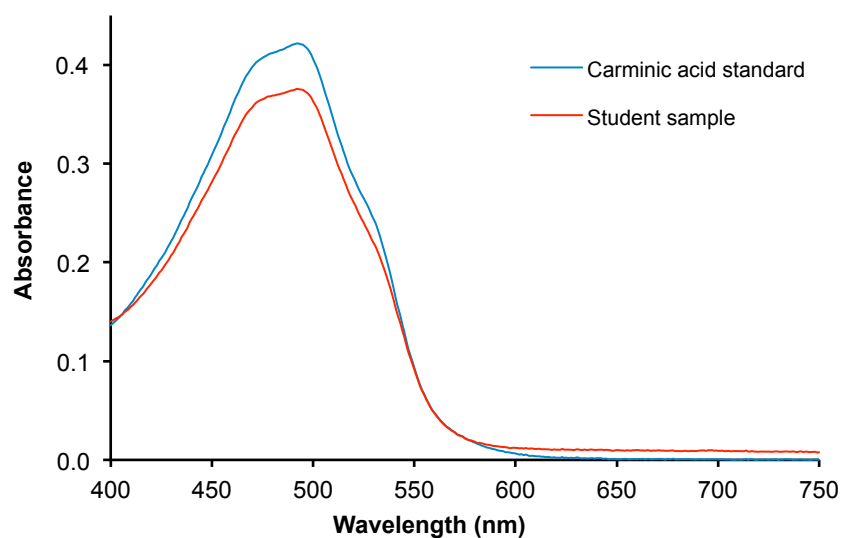


Figure 1. UV-Vis spectra of carminic acid: standard (blue); student sample (red).

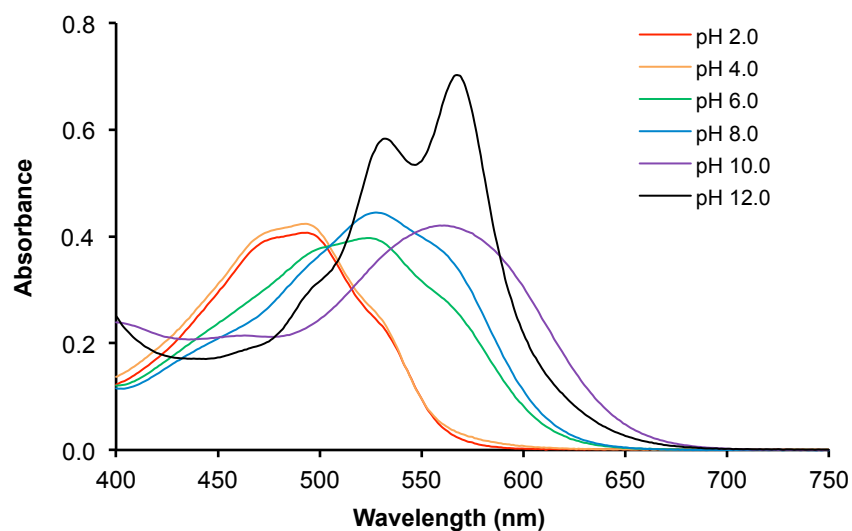


Figure 2. UV-Vis spectra of carminic acid at various pH.

λ/nm	222	279	316 (sh)	468 (sh)	493 (sh)	531 (sh)
$\epsilon/\text{L mol}^{-1} \text{cm}^{-1}$	7,200	12,200	4,100	2,600	2,700	1,700

Table 3. Tabulated values from quantitative UV-Vis spectra of carminic acid in water.¹

pH of Solution	2.0	4.0	6.0	8.0	10.0	12.0
$\lambda_{\text{max}} (\text{nm})$	492 ± 1	492 ± 1	523 ± 1	527 ± 1	558 ± 1	$566 \pm 1; 530 \pm 1$

Table 4. Absorbance maxima in visible region of carminic acid at various pH.

2.3 Stability of carminic acid as a function of pH

Carminic acid solution is stable between pH 2 and 10, but it is not stable when the pH is greater than 12 (compare to figure 2).

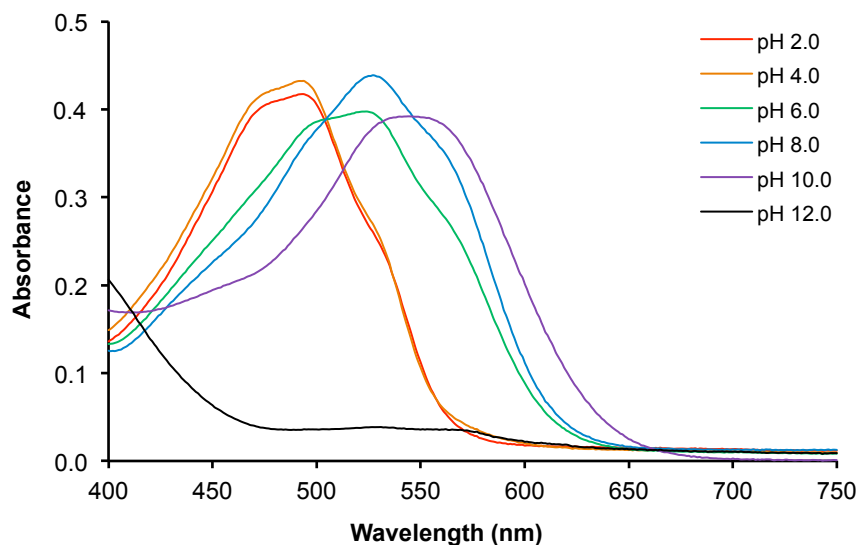


Figure 3. UV-Vis spectra of 50 μ M carminic acid solutions at various pH after 24 h.

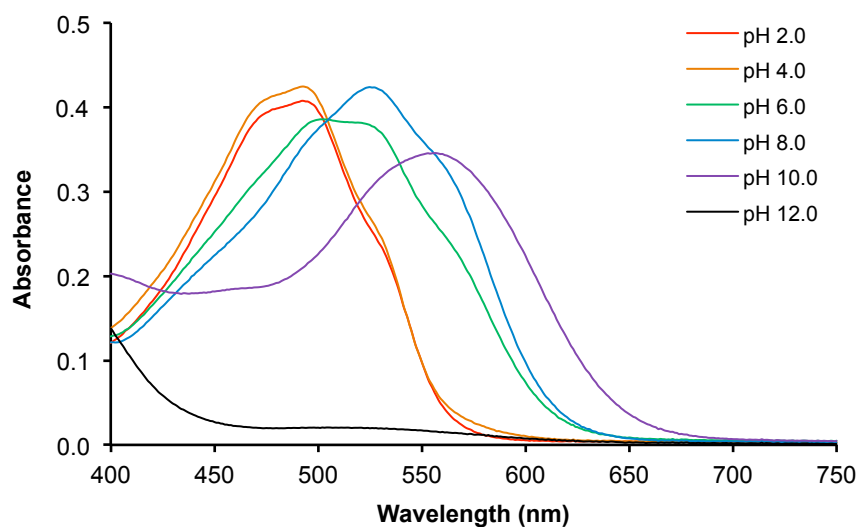


Figure 4. UV-Vis spectra of 50 μ M carminic acid at various pH after 48 h.

pH of solution	2.0	4.0	6.0	8.0	10.0	12.0
λ_{max} (nm)	492 $\pm$ 1	492 $\pm$ 1	523 $\pm$ 1	527 $\pm$ 1	558 $\pm$ 1	567 $\pm$ 1
Abs. (freshly made)	0.407	0.427	0.398	0.437	0.420	0.702
Abs. (24 h elapsed)	0.407	0.425	0.397	0.426	0.387	0.036
Abs. (48 h elapsed)	0.408	0.424	0.380	0.421	0.345	0.014

Table 5. Stability of 50 μ M carminic acid solutions at various pH.

2.4 FTIR spectroscopy

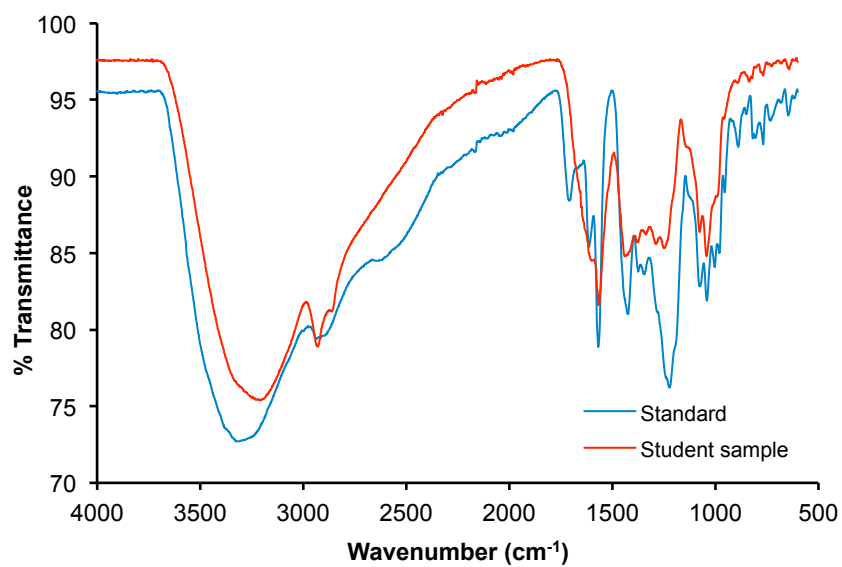


Figure 5. FTIR spectra of carminic acid: standard (blue); student sample (red).

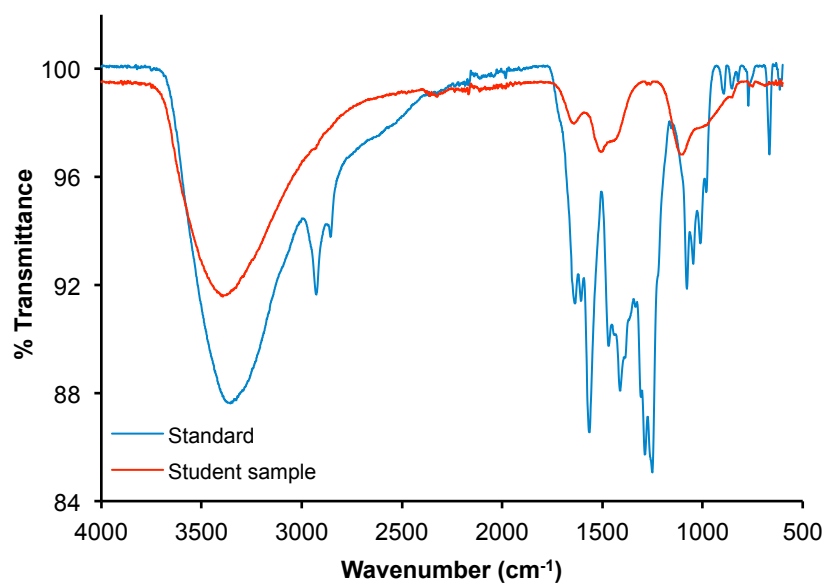


Figure 6. FTIR spectra of carmine: standard (blue); student sample (red).

2.5 Fluorescence characterization

The fluorescence spectrum of carminic acid is expected to have a $\lambda_{\text{max}} = 640$ and a shoulder at 750 nm when excited with 500 nm light.¹

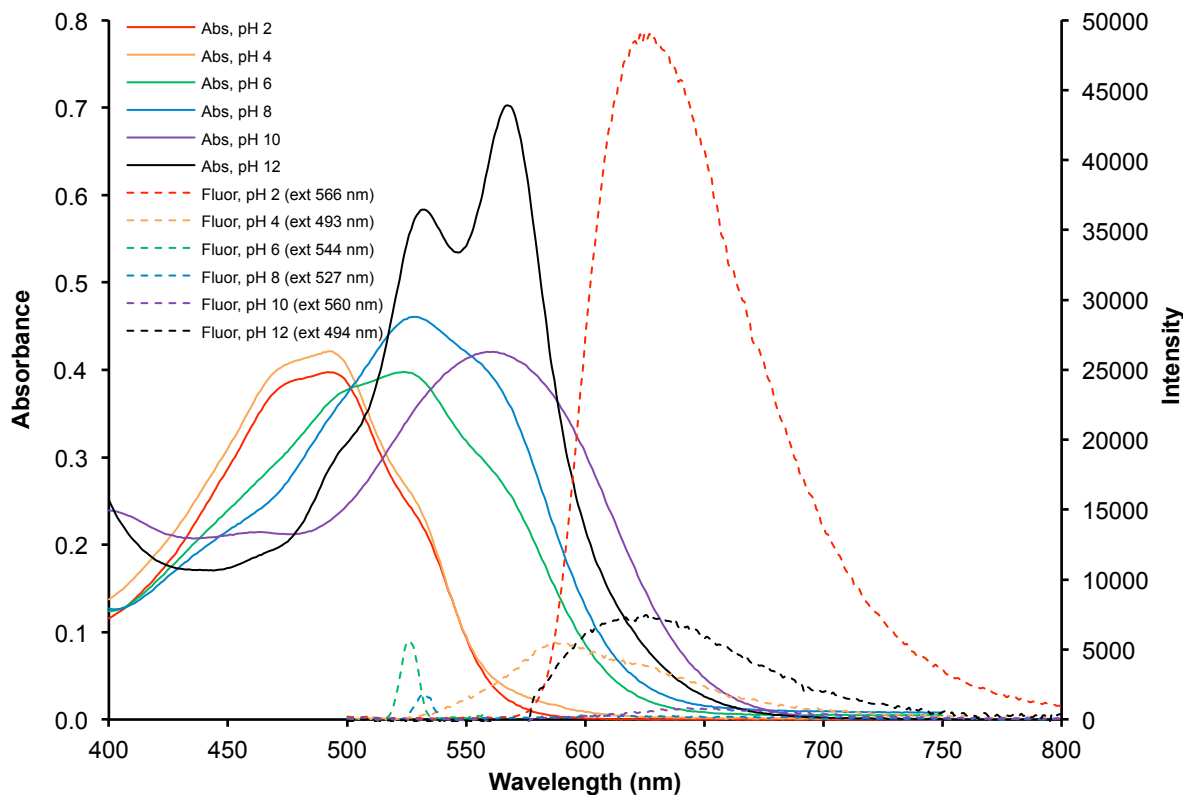


Figure 7. Fluorescence spectra (dotted lines) of 50 μM carminic acid overlayed with absorption spectra (solid lines) of the six species spectrally distinguishable when pH changes. All spectra were taken using Milli Q water as the reference blank.

pH of 50 μM carminic acid solution	Excitation wavelength (nm)	Emission wavelength region (nm)
2.0	492	500-800
4.0	492	500-800
6.0	523	500-800
8.0	527	500-800
10.0	558	500-800
12.0	567	500-800

Table 6. Fluorescence peaks for carminic acid.

2.6 Photoirradiation of carminic acid

A solution of carminic acid was prepared and irradiated with a 150 W Xenon lamp overnight. The absorbance change at 495 nm was monitored and the data fit to the equation for a first order reaction. A control sample was kept in the dark. Its absorption spectrum was measured at each time point.

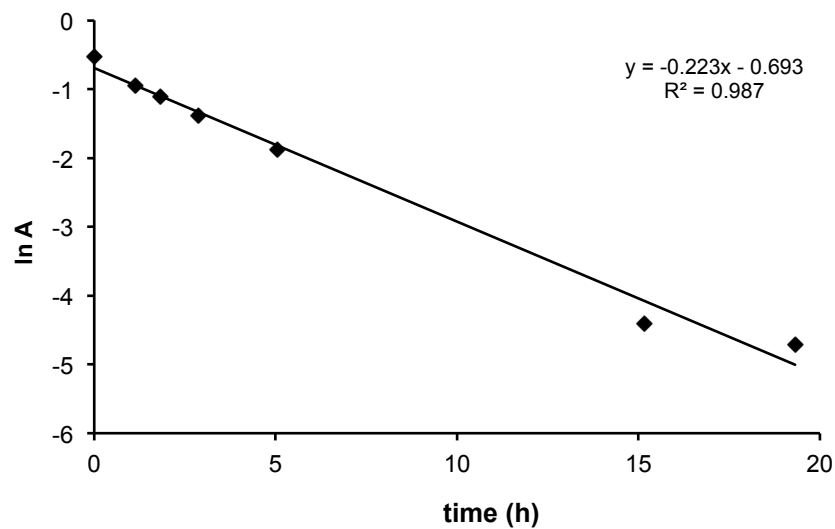


Figure 8. Plot of ln A vs. irradiation time for carminic acid. Slope, $m = -0.223 \pm 0.011$; y-intercept, $b = -0.69 \pm 0.11$ (using LINEST function in Excel).

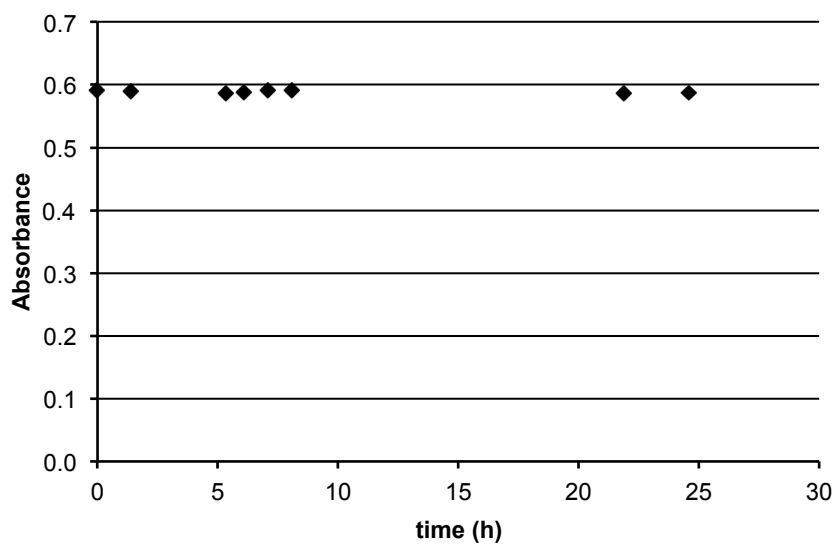


Figure 9. Absorbance vs. time for the dark control.

2.7 Cyclic Voltammetry

Solutions of carminic acid (100 - 400 μM) were prepared by dilution in pH 2.00 buffer (0.0106 M HCl with 0.05 M KCl). Cyclic voltammetric scans were recorded on a $\mu\text{Autolab}$ Type III potentiostat using commercially available glassy carbon working electrodes ($d = 3$ mm; BASi). Platinum wire ($d = 0.5$ mm.; Alfa Aesar) and Ag/AgCl (BASi) electrodes were used as the counter and reference, respectively. The glassy carbon electrode was mechanically polished with alumina suspensions (1 μm and 0.3 μm particle size; Buehler) in between measurements to minimize the extent of adsorption.

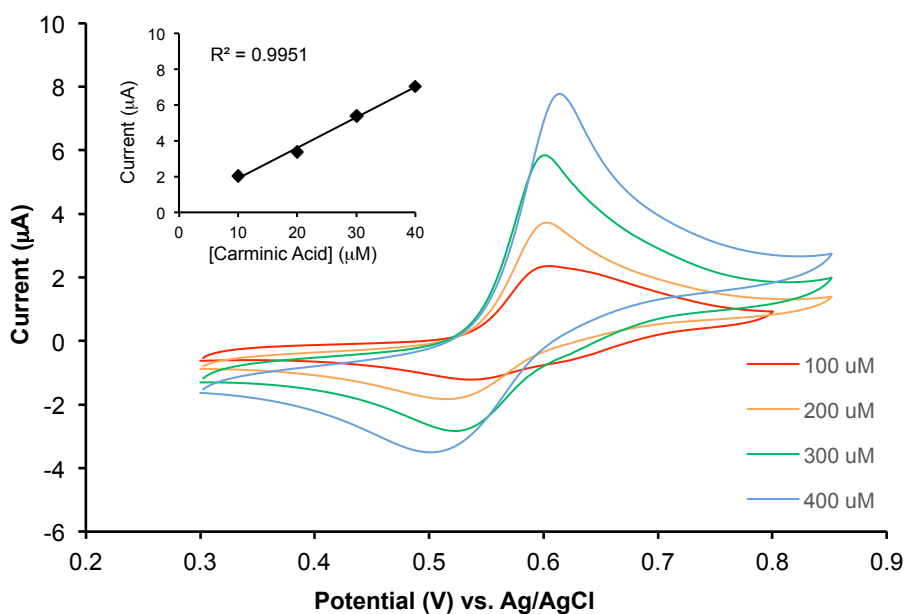


Figure 10. Cyclic voltammograms of carminic acid (100 – 400 μM) in pH 2 solution on a glassy carbon electrode (scan rate = 0.1 V/s). Scan 3 is plotted. Inset shows expected linear dependence of anodic peak currents on carminic acid concentration.

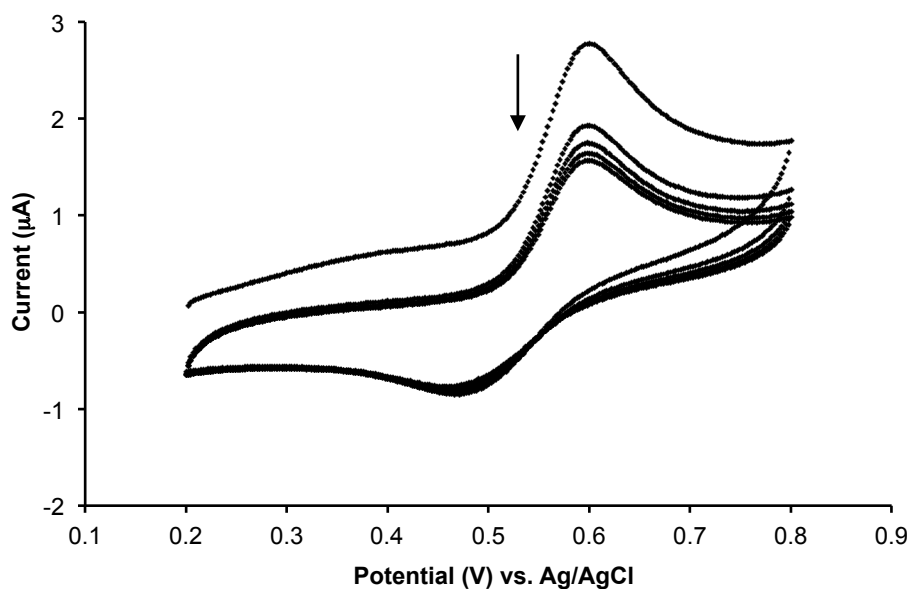


Figure 11. Consecutive cyclic voltammograms of carminic acid in pH 2 solution on a glassy carbon electrode (scan rate = 0.1 V/s). Anodic peak currents diminish with increasing scan number, indicative of surface adsorption.

2.8 Determination of Stoichiometry of Y^{3+} - carminic acid complex using Job's Method

In Job's method, the number of *total moles of metal + ligand* is kept the same throughout, while the *ratio of metal to ligand* is varied across a series of solutions. In our experiment, we used solutions of carminic acid and yttrium that were equimolar in concentration to two significant figures, so that we could simply vary the volumes of carminic acid and yttrium instead.

The absorbance at the λ_{max} of the complex being formed (which is directly proportional to concentration of complex) vs. mole fraction of ligand was plotted for each solution. The resulting plot look like an upside-down V: at a certain mole fraction of ligand, the absorbance reaches a maximum; this represents the point at which neither metal nor ligand is in excess and the maximum concentration of complex is reached. At this point, the ratio of ligand to metal is exactly equal to the stoichiometry of the complex formed. We monitored the change of absorbance at 573 nm.

The best way to determine the vertex of the V is to fit each side to a linear equation and then determine the point of intersection by setting the equations equal to each other. Linear fits of the yttrium(III)-carminic acid complex were determined using the trendline feature in Excel. They intersect when the mole fraction of carminic acid = 0.5, indicating that Y^{3+} and carminic acid are present in a 1:1 ratio in the complex.

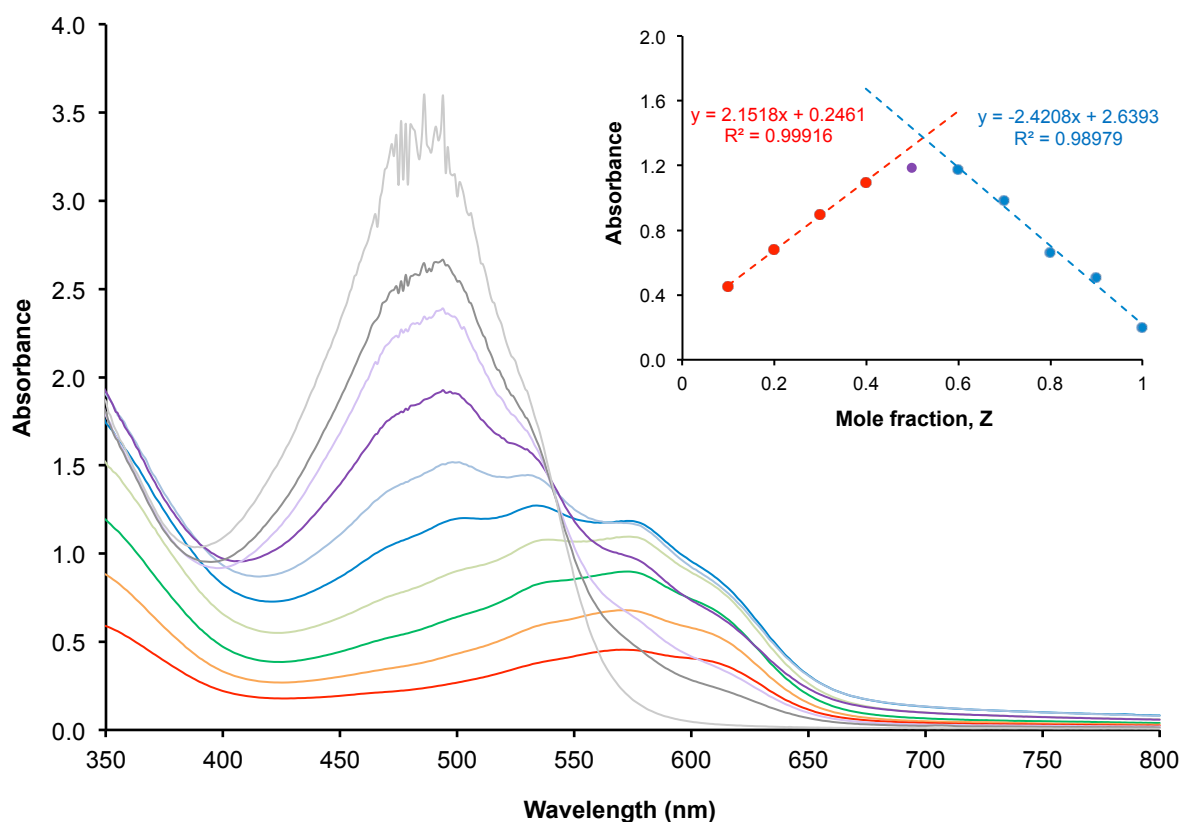


Figure 12. Plot of wavelength vs. absorbance during yttrium titration of carminic acid. Inset: Job's analysis of Yttrium carminic acid titration.

3. Hazards

These experiments present few hazards to the student. Students should use appropriate PPE at all times, which includes wearing lab coats, goggles, and gloves. Appropriate laboratory attire is expected, which include clothes that leave no exposed skin, closed toed shoes, securing long hair, and wearing no dangling jewelry or scarves. Acids and bases should be handled appropriately. A hot plate is used to extract the carminic acid and students should use it with care. The UV lamp should be used with safety goggles rated for UV light and students should not look directly at the lamp. Students should read instructions and be trained on all instruments before operating.

III. Evaluation of student outcomes

The goal of this experiment was to stimulate students to draw on material they had used earlier in the course or in other courses to design and carry out experiments associated with the general task of characterizing the electronic structure of carminic acid and carmine. In our case, the lab was the only open-ended experiment in an otherwise highly structured set of spectroscopic experiments. We view this as a strength because students had already been trained on how to use various pieces of equipment in our department and how to use the software needed to analyze the resultant data but can imagine other models in which this laboratory would work well.

In evaluating student outcomes we asked students to describe what they felt they learned from doing this lab and how it compared to what they experienced when they did other experiments every step of the experimental protocol was given to them. Here are the relevant responses (student comments in italics).

The project overall was very much a detective job, which reminded me of "detecting unknown mixture" experiment in Modern Techniques in Organic Chemistry class, whereby students were given an unknown mixture of acidic, basic, or neutral compound (some had liquid mixture, some had liquid-solid etc.) and had to chemically separate, purify and analyze using NMR to find out the content. It made me think about the materials from organic, inorganic and analytical chemistry materials. Even though we couldn't include it in the paper, we did consider kinetics experiment to measure the rate of deprotonation as well.

It gave us a condensed version of start-up research experience. Even though many of us have had substantial research experience by the time we started the project, it was very useful because not all of us get to start a completely new project (I think it is more common for an undergrad student to take over the previous lab member's project). It is a unique experience to be starting a brand-new project where even the instructors are still figuring out what can be done.

I thought it was a great way to test our skills that we've acquired in the earlier lab courses, especially because it is easy to confuse "knowing how to operate the instrument" vs. "understanding the fundamentals of the instrument and utilize it to your purpose", particularly for spectroscopy-based labs. Unlike wet labs that we get more exposure since freshmen year, spectroscopy labs were fairly new to many of us (except for those who routinely use them in their research) as well.

I can understand if some students preferred the "cookbook" style, but I learnt from my research experience that it is not always the case in real life. When confronted with uncertainty, what we can rely on as chemists are 1) a solid understanding of previous works to develop a good (or complementary) experimental design, and 2) an intelligent choice of analytical tools. The literature search, class discussion, and experiments all helped us strengthened these skills. Perhaps, we could have used longer preparation period before taking the measurements, but I think we were able to obtain promising results for future projects.

I don't like "cut and dried" experiments where we just follow a set procedure and analyze the data. It's great to look at a simple system and see that certain spectroscopy techniques give different pieces of data, but I think being a researcher/scientist is to apply the techniques we learn to answer questions we care about. I think having the carminic acid project was a test of how much we learned--in other words, it was our final exam. And the open-ended lab was still a learning process because in science, there are challenges that crop up as data is collected.

I definitely found myself thinking about material from other courses with the carmine lab--as we were thinking about where the carminic acid might oxidize first and when we were discussing where the carminic acid could be bound to a metal. Having to integrate ideas from multiple courses I had taken was a really nice culmination of my time in chemistry at Barnard.

I felt like I gained more from this lab in terms of actually learning how to use the instruments by being able to go back to techniques from earlier in the semester another time, especially fluorimetry which I had never used before.

Though I personally didn't do those experiments, I thought it was really cool to have electrochemistry and photodegradation (photostability?) experiments (what Jackie was doing) incorporated into a lab course. I've come across various papers in the literature in the past year related to these topics in the more organic chemistry related papers that I've read, and it's cool that we had the opportunity to think about/use them in a lab course setting. I feel more inclined to ask people about their chemistry related to dyes and have talked with people casually here about things related to the carmine lab.

Ultimately I think having a more open ended lab like this one is more fun and interesting to think about than set planned labs, particularly as senior/junior chem majors.

All of the negative comments from students were about the extra time that the more open-ended experiments took, in part because of the emotional energy required to think through this project at the same time that many of them were trying to finish their senior theses. We were not able to offer students unfettered access to the lab and had to restrict them to times when they would be supervised, which also exacerbated the sense of time pressure.

It is a tough call. If this lab continues to be scheduled in spring semester, it can be difficult to have an open-ended lab like this semester, as seniors have senior thesis/independent research project coming up. I found it difficult to put more than necessary effort in carmine project, because I wanted to work more on my thesis. However, it was very rewarding and exciting experience to start everything from scratch as a team.

I liked the flexibility of the carminic acid lab and the discussions that came with it, but I think it would be most fulfilling to have more time dedicated to a lab set up in a more open-ended way.

References

1. Kunkely, H.; Vogler, A. *Inorganic Chemistry Communications* **2011**, 14 (7), 1153-1155.
2. Borges, M. E.; Tejera, R. L.; Díaz, L.; Esparza, P.; Ibáñez, E. *Food Chemistry* **2012**, 132, 1855-1860.